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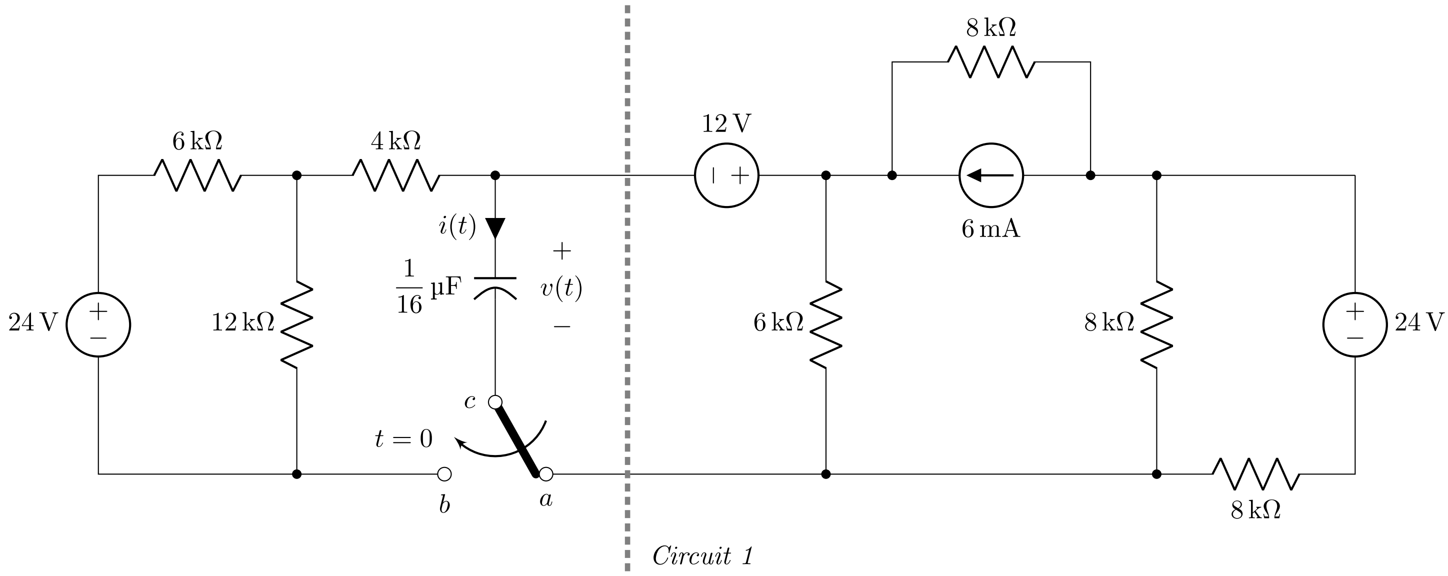
Brac University
Semester: Fall 2025
Course Code: CSE250
Circuits and Electronics

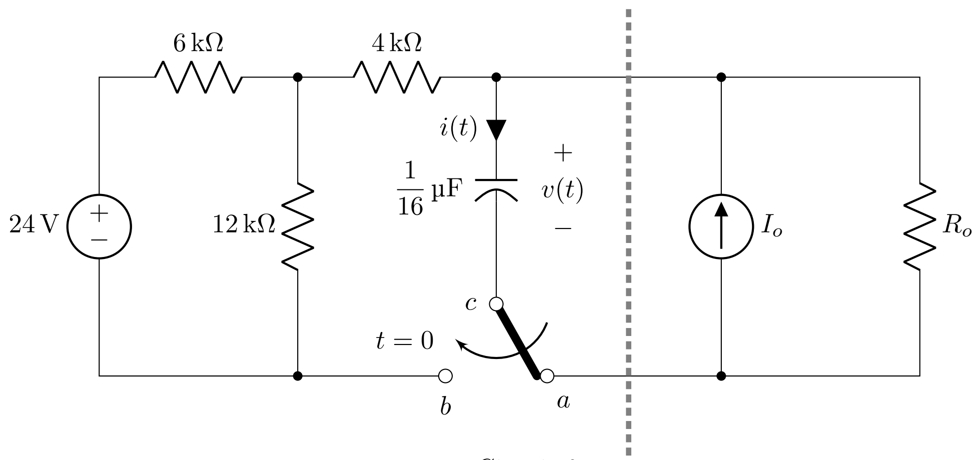
Assessment: *Final Exam*
Duration: 1 hour 40 minutes
Date: February 11, 2026
Full Marks: 55

- ✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ At the end of the exam, both the **answer script** and the **question paper** must be returned to the invigilator.
- ✓ All **3 questions** are compulsory. Marks allotted for each question are mentioned beside each question.
- ✓ Proper units must be included for all calculated values. Marks will be deducted for missing or incorrect units.
- ✓ Symbols have their usual meanings.

■ Question 1 of 3 *[CO3] [20 marks]*



Circuit 1



Circuit 2

Analyze the circuits shown above to answer the following questions:

(a) *[8 marks]* Reduce the portion of *Circuit 1* to the right of the vertical dashed line† so that it takes the form of *Circuit 2*. Hence, determine the values of I_o and R_o.

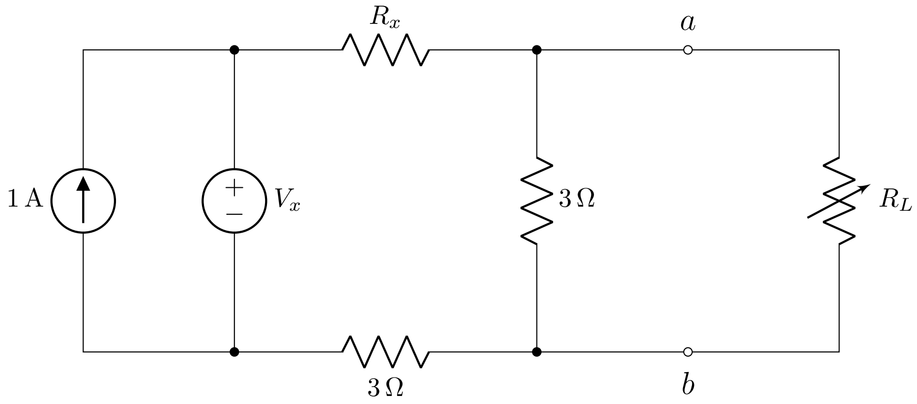
(b) *[9 marks]* Determine the voltage $v(t)$ across the capacitor as a function of time t, for $t > 0$.

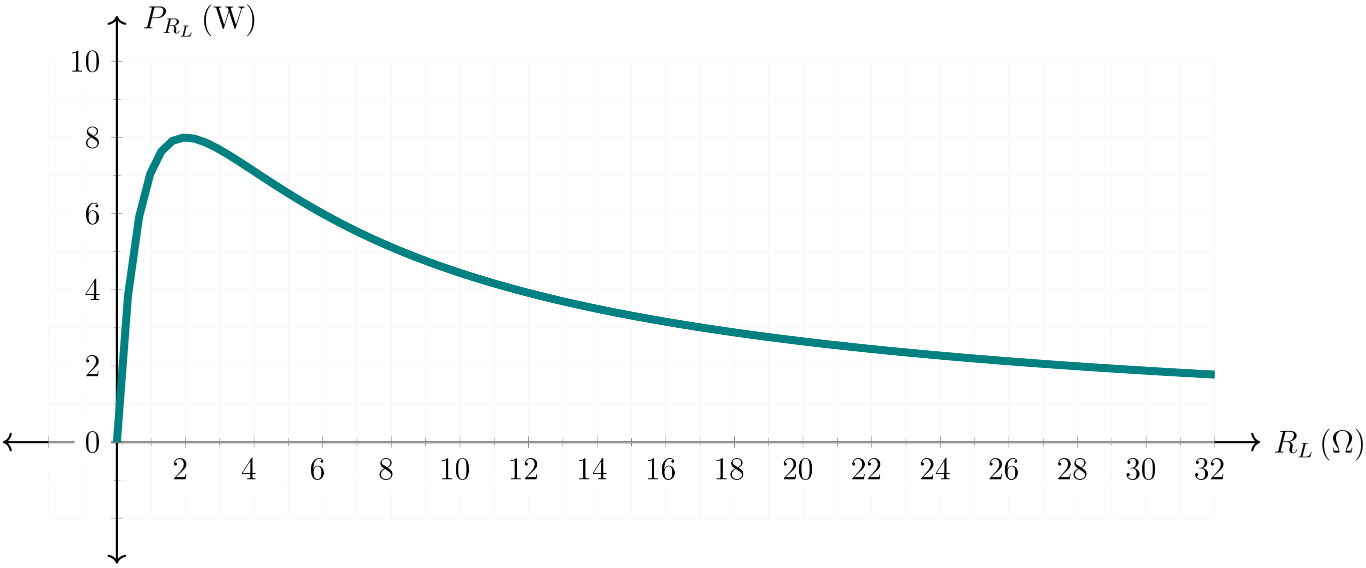
(c) *[3 marks]* Determine the capacitor current $i(t)$ as a function of time, for $t > 0$.

†The vertical dashed line indicates a boundary for circuit reduction and does not represent any circuit element or physical connection.

■ Question 2 of 3 *[CO2] [23 marks]*

(a) In the following circuit, the power delivered to the resistor R_L is plotted below as R_L is varied.

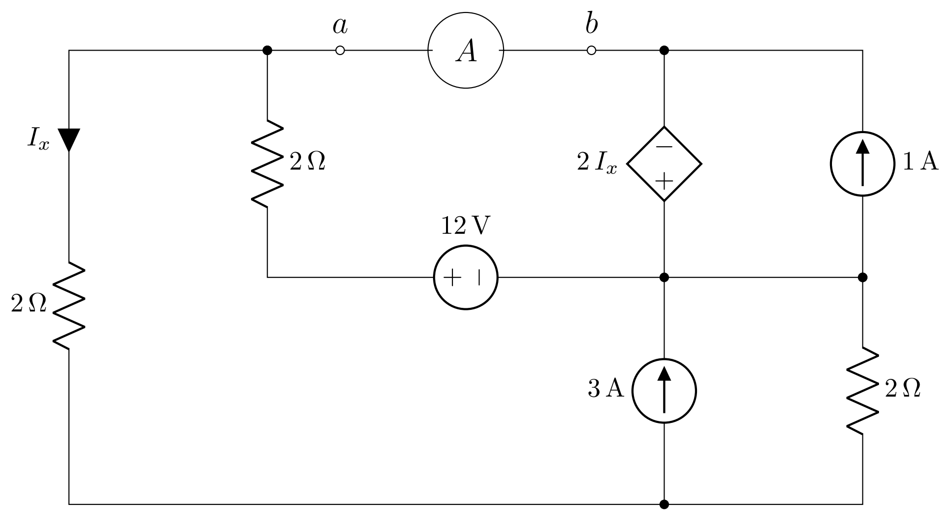




(i) *[4 marks]* Determine the unknown resistance R_x in the circuit.

(ii) *[4 marks]* Determine the unknown voltage V_x.

(b) The ammeter A in the following circuit is used to measure the current flowing between terminals a and b.



Apply Thevenin's Theorem or Norton's Theorem to answer the following questions:

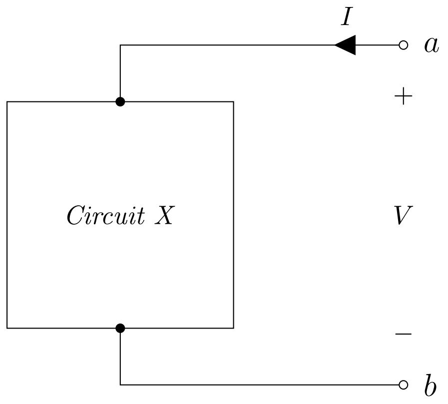
(i) *[12 marks]* Determine the Thevenin or Norton equivalent circuit with respect to terminals a and b.

(ii) *[3 marks]* If the ammeter can safely carry a maximum current of 1 A, use the equivalent circuit obtained in (i) to determine the minimum value of resistance of the ammeter required so that it operates safely without exceeding the maximum allowable current.

■ Question 3 of 3 *[CO3] [12 marks]*

(a) *[4 marks]* X is a linear two-terminal circuit connected between terminals a and b, as shown below. When tested, the circuit produced the voltage V and current I values listed in the adjacent table; however, two entries are missing.

Calculate the missing values and complete the table.



V (volt)	2	6	10	
I (A)	−0.5	0.5		2.0

(b) *[8 marks]* Determine whether the following circuit is equivalent to *Circuit X* in (a).